Daniele Ballinari

Quantitative researcher with advanced data science skills and a passion for causal machine learning and time series econometrics.

Education

University of St. Gallen, Switzerland

Ph.D. in Economics and Finance (Grade: 6/6, prize for best Ph.D. thesis) *07/2018 — 11/2020*
- Dissertation: *Essays on investors' sentiment and attention*, under the supervision of Prof. Francesco Audrino and Prof. Zhi Da.

M.A. in Quantitative Economics and Finance (Grade: 5.71/6) *09/2016 — 06/2018*
- Thesis: *Non-Linear Effects of Economic, Sentiment and Attention Measures on Realized Volatility.*

B.A. in Economics (Grade: 5.79/6) *09/2012 — 06/2016*
- Thesis: *Optimal hedge ratio for Swiss pension funds.*

Relevant Work Experience

Lead Data Scientist – Technology and Data Science, Money Market and Foreign Exchange, Swiss National Bank *04/2025 — Now*
- Leading the team's portfolio of data science solutions, products and projects for the implementation of monetary policy.
- Fostering the application and the knowledge of machine learning and AI solutions across the Swiss National Bank.

Data Scientist – Technology and Data Science, Money Market and Foreign Exchange, Swiss National Bank *12/2021 — 03/2025*
- Product owner and developer of econometric and machine learning applications supporting monetary policy implementation.
- implementation of AI solutions to process textual data for market intelligence and sentiment analysis.

Lecturer — University of Basel & University of St. Gallen *09/2021 — 12/2024*
- University of St. Gallen: Lecturing the Master's course *Asset Pricing* and the Bachelor's course *Mathematics for Economics.*
- University of Basel: Lecturing the Bachelor's course *Introduction to Time Series Econometrics* and the Master's course *Univariate Time Series Analysis.*

Postdoctoral researcher — University of Basel *01/2021 — 11/2021*
- Conducting research on the project *Sustainable Finance and Investor Sentiment* in collaboration with Prof. Mahmoud.
- Applying state-of-the-art natural language techniques to identify opinions on sustainability from social media data.

Asset-Liability Management Consultant — c-alm AG, St. Gallen *01/2021 — 11/2021*
- Project-based consulting for pension funds, analyzing assets and liabilities to enhance stability.
- Provide strategic insights to optimize sustainability, investment strategies, and regulatory alignment.

Selected Academic Research

The impact of sentiment and attention measures on stock market volatility
- Analysis of the impact of sentiment and attention variables on stock market volatility using a novel and extensive dataset that combines social media, news articles, information consumption, and search engine data.
- Joint work with Francesco Audrino and Fabio Sigrist, published in *International Journal of Forecasting* (2020).

When does attention matter? The effect of investor attention on stock market volatility around news releases
- Combining social media and news article data, the analysis uncovers how retail and institutional investor attention is related to the way stock markets process information.
- Joint work with Francesco Audrino and Fabio Sigrist, published in *International Review of Financial Analysis* (2022).

Calibrating doubly-robust estimators with unbalanced treatment assignment
- Finite-sample improvement with asymptotic guarantees for machine learning-based average treatment effect estimators in the presence of unbalanced treatment assignment.
- Published in *Economics Letters* (2024).

Semiparametric inference for impulse response functions using double/debiased machine learning
- Introduce and provide theoretical results for a double/debiased machine learning estimator for the impulse response function in settings where a time series of interest is subjected to multiple discrete treatments.
- Working paper with Alexander Wehrli (2024).

Improving the finite sample estimation of average treatment effects using double/debiased machine learning with propensity score calibration
- Probability calibration to improve machine learning-based average treatment effect estimators.
- Working paper with Nora Bearth (2024).

Relevant Skills & Expertise

- **Advanced Technical Skills**: Python, Spark, Torch, SQL, R, Git, Latex, Matlab.
- **Languages**: Italian & German (bilingual), English (proficiency), French (beginner).
- **Certificates**: TensorFlow Developer Certificate, Professional Scrum Master, ACI Dealing Certificate.
- **Grants and Awards**: Prize for best Ph.D. Thesis in Economics and Finance (2021), WWZ Förderverein Funding (2021), Swiss National Science Foundation Grant (2017-2020).